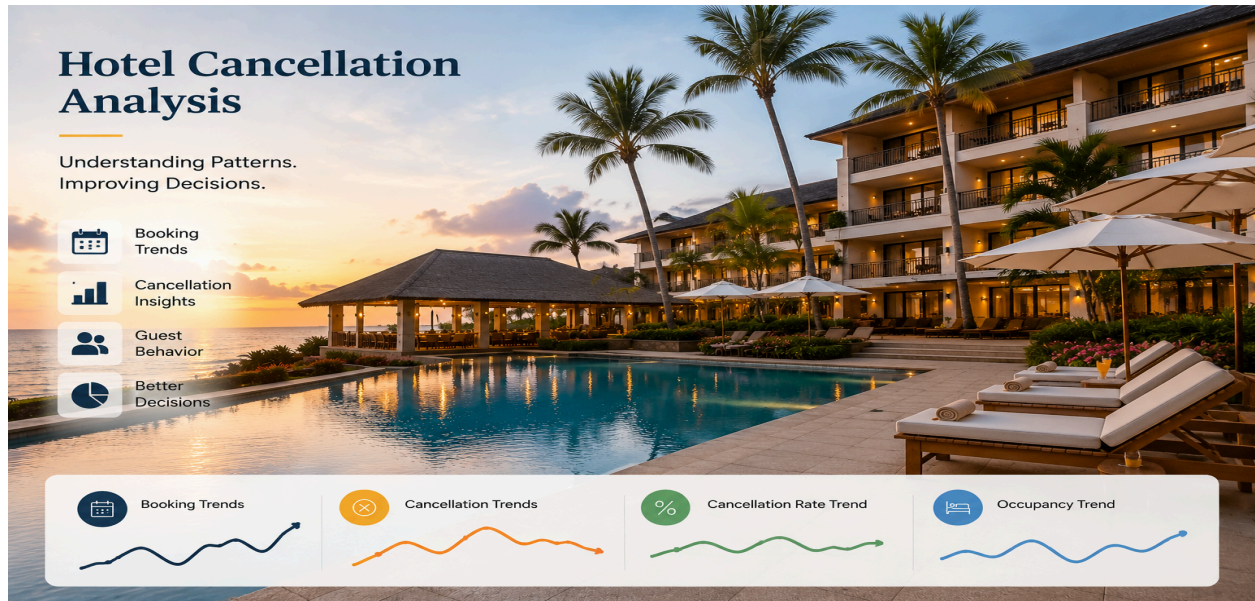


Hotel Cancellation Analysis Project Report



Hotel Booking Cancellation Analysis

- By Sourabh Biradar

1. Introduction

Hotel booking cancellations are a major challenge for the hospitality industry as they directly impact revenue generation, occupancy forecasting, resource allocation, and operational planning. Understanding the factors contributing to cancellations can help hotels make better strategic and pricing decisions.

This project focuses on analyzing hotel booking data to identify cancellation trends, customer behavior patterns, and business insights using Python-based data analysis techniques.

2. Problem Statement

Hotel booking cancellations create uncertainty in demand forecasting and revenue management. The objective of this project is to analyze booking data and identify the factors that contribute to cancellations.

3. Objectives

- Analyze hotel booking cancellation patterns
 - Identify important factors affecting cancellations
 - Perform data cleaning and preprocessing
 - Conduct exploratory data analysis (EDA)
 - Generate business insights and recommendations
-

4. Dataset Information

The dataset contains hotel booking information such as:

- Hotel type
- Booking status
- Country
- Lead time
- ADR (Average Daily Rate)
- Reservation dates
- Customer details
- Number of guests
- Market segment

Dataset Overview

- Rows: 119390 records
 - Columns: 32 features
-

5. Tools & Technologies Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

6. Data Wrangling / Data Cleaning

Several preprocessing steps were performed to improve data quality and prepare the dataset for analysis.

6.1 Loading Dataset

The dataset was loaded using pandas.

```
import pandas as pd
df = pd.read_csv("../data/raw_data.csv")
```

6.2 Dataset Inspection

Basic checks were performed:

- Shape of dataset : df.shape
 - Column names : df.columns
 - Basic info : df.info()
 - Summary statistics : df.describe()
-

6.3 Datetime Conversion

The reservation status date column was converted into datetime format.

```
df['reservation_date'] = pd.to_datetime(
    df['reservation_status_date'],
    format='%d/%m/%Y',
    errors='coerce')
```

6.4 Dropping Unnecessary Columns

The original reservation status date column was removed after conversion.

```
df = df.drop(columns='reservation_status_date')
```

Columns with excessive missing values such as:

- agent
- company

were also removed.

```
df = df.drop(columns=['agent', 'company'])
```

6.5 Missing Value Handling

Missing values were identified using:

```
df.isna().sum()
```

I dropped 'agent' & 'company' columns since they had too many missing values , way beyond what we can handle.

6.6 Duplicate Check

Duplicate rows were identified and analyzed to ensure data quality.

```
df.duplicated().sum()
```

7. Exploratory Data Analysis (EDA)

EDA was conducted to identify patterns, relationships, and trends in hotel booking cancellations.

7.1 Cancellation Analysis

Cancelled bookings were filtered using:

```
cancelled = df[df['is_canceled'] == 1]
```

This helped analyze cancellation-specific behavior.

7.2 Country-wise Cancellation Trends

Top countries with the highest cancellation rates were identified.

```
top_10 = cancelled['country'].value_counts()[:10]
```

Insights :

Since the hotels are based in Portugal, local bookings and cancellations were disproportionately high; therefore, Portugal was excluded to avoid bias in the analysis.

7.3 Booking Trends

Booking distributions and customer behavior were analyzed using:

- Count plots
 - Pie charts
 - Line plots
-

7.4 Outlier Analysis

Boxplots were used to detect outliers on `df['adr']` & IQR and also handled outliers using

```
df['adr'] = df['adr'].clip(lower,upper)
```

8. Key Insights

Insight 1

Hotels experienced around 37% of booking cancellations.

Insight 2

City Hotel experienced a significant number of booking cancellations.

Insight 3

Higher Avg Daily Rate ('adr') were associated with higher cancellation probabilities.

Insight 4

GBR: United Kingdom (Great Britain), **ESP:** Spain, **FRA:** France, **ITA:** Italy, **DEU:** Germany

Were top 5 countries with the most number of cancellations. (Excluding Portugal)

Insight 4

Customer booking behavior differed across hotel types.

Insight 5

Seasonal booking patterns impacted cancellation trends.

9. Business Recommendations

Based on the analysis, the following recommendations were proposed:

- Reduce long free-cancellation windows
 - Implement dynamic pricing strategies
 - Improve customer engagement before check-in
 - Monitor high-risk booking segments
 - Introduce cancellation prediction systems
-

10. Challenges Faced

- Handling missing values
 - Managing inconsistent datetime formats
 - Identifying meaningful duplicates
 - Cleaning categorical inconsistencies
-

11. Conclusion

This project successfully analyzed hotel booking cancellations using Python and data analysis libraries. Through data cleaning, preprocessing, and exploratory data analysis, several important trends and customer behavior patterns were identified.

The analysis can help hotel businesses improve operational planning, optimize pricing strategies, and reduce revenue loss due to cancellations.
